

**REQWEST: A BARANGAY DOCUMENT REQUEST SYSTEM
FOR THE RESIDENTS OF WEST KAMIAS,
QUEZON CITY**

A Capstone Project
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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

Barangays are the Philippines' smallest governmental units, and they contribute to ensure the safety of its citizens. Barangays are also in charge of providing and disseminating vital official papers, such as clearances and certificates, as requested by citizens. Taruc et al. (2023) found that barangays that rely on manual document request processing face several issues, including difficulties in managing files and records, ensuring file security due to the absence of a secure repository, and generating barangay records in real-time.

In many cases, barangay residents have a hard time processing and requesting their documents in their barangay. Some are having trouble queueing a long line and residents being confused on the requirements needed. Common issues include delayed responses from barangay officials, difficulties in managing and retrieving files, and a lack of organized record-keeping.

According to researchers from Olongapo City University (2021) Residents in various barangays have reported challenges in obtaining necessary documents due to inefficiencies in manual processing systems.

Having a Web-based Document request system can help the barangay administrators to track and compile the requirements of the residents and help the residents to have their request submitted online and waiting for them to pick it up.

REQWEST is designed to alleviate the difficulties residents face in obtaining necessary Documents, promoting a more efficient and responsive barangay administration.

Theoretical Framework

This study is grounded on two key theories: the Technology Acceptance Model (TAM) and the Information Systems Success Model (ISSM). These theories support the development and evaluation of the REQWEST system in terms of usability, functionality, and user acceptance.

1. Technology Acceptance Model (TAM) Developed by Davis (1989), the Technology Acceptance Model explains how users come to accept and use a technology. It highlights two main factors:

- Perceived Usefulness (PU) – the degree to which a person believes that using the system would enhance their performance or productivity.
- Perceived Ease of Use (PEOU) – the degree to which a person believes that using the system would be free of effort.

In the context of REQWEST, residents and barangay officials are more likely to adopt the system if they find it useful for requesting and processing documents and easy to navigate even with minimal technical skills.

2. Information Systems Success Model (ISSM) Proposed by DeLone and McLean (1992), the ISSM outlines several dimensions that determine the success of an information system:

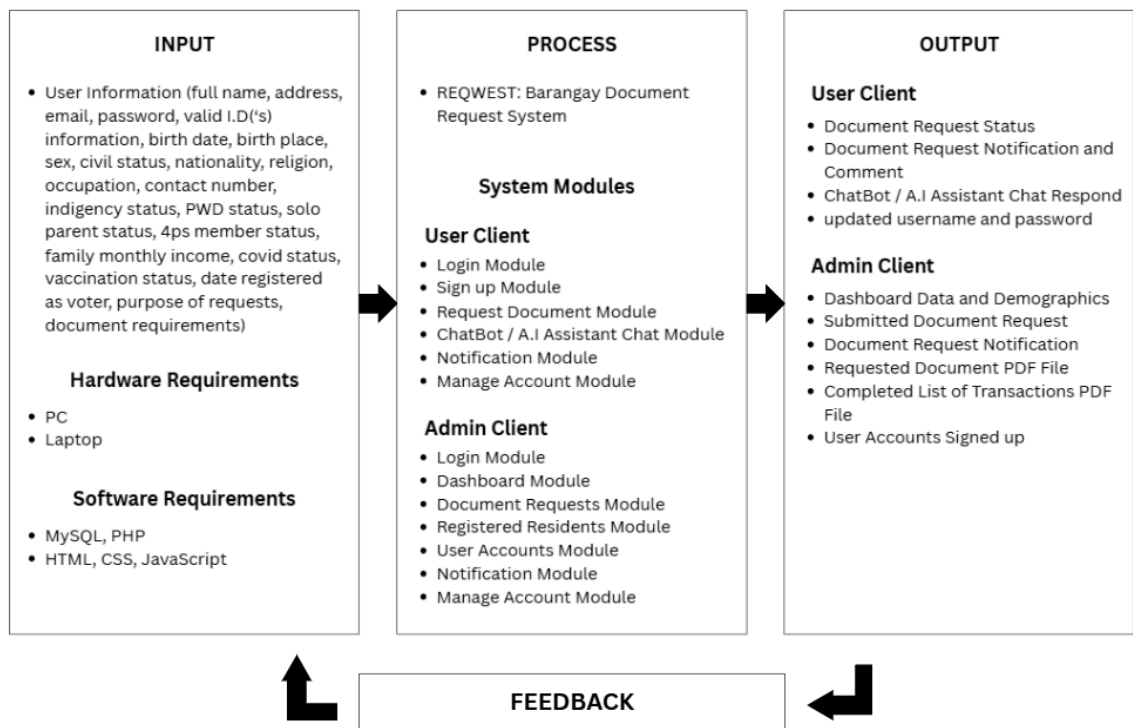
- System Quality – performance, reliability, and ease of use.
- Information Quality – accuracy and relevance of information.

- Service Quality – support and responsiveness.
- User Satisfaction and Net Benefits – impact on individual or organizational performance.

This model justifies the evaluation criteria used in the study such as functional suitability, performance efficiency, interaction capability, reliability, flexibility, and security, all of which are essential to measuring the effectiveness and success of the REQUEST system.

Conceptual Framework

Figure 1. Conceptual Framework



Statement of the Problem

The current process of requesting documents from the barangay office is often time-consuming and inefficient, leading to inconvenience for residents. Despite the growing need for digital solutions, many barangay offices still rely on manual systems

that lack accessibility, real-time tracking, and user-centered features. To address these issues, this study aims to develop a Barangay Document Request System (REQWEST) that simplifies and streamlines the process for residents

This study seeks to answer the following questions:

1. What is the profile of the respondents in terms of:
 - 1.1 Age;
 - 1.2 Gender;
 - 1.3 Senior citizen and PWD Status
 - 1.4 Employment status
2. What type of document is the most frequently requested by residents from the barangay?
3. What is the level of user acceptance of REQWEST: A Web-based Document Request System for the Residents of Barangay West Kamias in terms of:
 - a. Functional Suitability
 - b. Performance Efficiency
 - c. Interaction Capability
 - d. Reliability
 - e. Flexibility
 - f. Security

Scope and Limitation of the Study

This study is focused on the "REQWEST – Barangay Document Request system for the residents of West Kamias". The researchers focus on developing a centralized web-based system that will facilitate barangay document requests. The system will allow

residents to submit the requests online, track the status of their applications, and

However, The system will be limited to Barangay Kamias, Quezon City, and will not support document requests from other Barangay areas.

REQWEST operates through two (2) authorized system users: barangay officials and local residents. The system only includes the following:

a. A system-based document authentication cannot process any requests for documents from unregistered residents.

b. Individuals who are not capable of using the internet will not access barangay online functionality, so individuals must submit their requests by visiting the barangay office physically.

c. The users are only those who are officially registered as voters within Barangay West Kamias.

d. The only documents available for request through the system are the following:

- Barangay Certificate of Residency
- Barangay Permit (for school, travel, and employment purposes)
- Barangay Certificate of Good Moral Character
- Barangay Certificate of Indigency

e. When submitting a request for a document, only the softcopy file given by the residents is acceptable as supporting evidence.

f. Although REQWEST facilitates the online submission and processing of requests, residents are still required to claim their approved documents in person at the barangay office.

g. The system is designed for essential functionalities only. It does not include

advanced features such as online payment transactions, which may be considered for enhancement in future system upgrades and for further research.

Significance of the Study

This study would be beneficial to the following:

1. Barangay Officials – Officials of Barangay West Kamias will benefit from having better access to and management of resident information and document requests. The system will streamline administrative processes, improving efficiency in handling service requests.
2. Citizen (Brgy. West Kamias) – Citizens will have easier access to request documents such as birth certificates (PSA), barangay clearances, voter's IDs, barangay IDs, indigency certificates, and business permits. The system also prioritizes certain groups, including:
 - 2.1 Senior Citizen and Person with Disability (PWD) - These individuals will receive priority service and free service as stated by the barangay secretary.
 - 2.2 Student – Students also receive free services.
3. Future Researchers – Future researchers will benefit from this study as it provides a foundation for further development. They can improve and refine the existing system to address emerging needs and enhance its functionality.

Definition of Terms

Document Management System. This term refers to a system that tracks and stores documents digitally to enable easier access and retrieval.

System Implementation. The process of installing and integrating a system into an operational environment, such as local barangays.

Public Service. Services provided by the government to meet the needs of the public.

Community Engagement. The process of involving community members in planning, and execution of projects or services.

Data Management. The processes used to collect, store, and organize data for accessibility.

Web-based System. A system or application that runs on a web browser, accessible via the internet.

Priority Groups. Groups that are given priority in receiving services or benefits due to specific needs, such as senior citizens or persons with disabilities (PWDs).

User Interface. A digital surface in which a user interacts with a computer or software, including the layout, buttons, and controls on a screen.

Record-Keeping. The practice of systematically maintaining and storing records for reference, legal, or administrative purposes.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

To validate the use of a web-based portal as a potential document request system, the following literatures have been provided to further expand the knowledge and explain the purpose as to why the study will be conducted. Only those that were relevant were studied.

Barangay Information Management System

As the smallest political unit in the Philippines' governmental structure, the barangay is responsible for the fundamental platform for local administration. Most barangay operations today depend on manual techniques for record management beside information processing. Taruc et al. (2023) found that barangays that rely on manual document request processing face several issues, including difficulties in managing files and records, ensuring file security due to the absence of a secure repository, and generating barangay records in real-time. Clearances and certifications are prepared with less to no base-line data to assure reliability. This opens the risk for nonresidents to take due advantage of this process flaw (Ong, 2020). As stated by Arcega et. al (2023), Barangay administration has document and record management as an essential duty since these records support both legal documents and certification needs as well as residency verification. The essential management of documents and records are available to facilitate various purposes such as barangay clearance and certificate generation alongside ID production and proof of residency and other barangay documents.

Barangay Management Information Systems now dominate as people depend more intensely on computer systems across all aspects of life, especially in business operations. By guiding business day-to-day operations, the productivity of workers increases because they spend less time on data collection and more time addressing essential work functions. (Maribao et. al, 2024)

Accordingly, residents formerly needed to visit the barangay hall, complete paperwork, and then have to wait extended processing time to obtain these documents from the local office. The development of technology together with growing public service effectiveness demands has led numerous barangays to implement the Barangay Document Request System.

In a study conducted by Santiago et al. (2023), With technological advancements and the growing demand for more efficient public services, many barangays have started to implement digital solutions to simplify administrative processes. One such innovation is the Barangay Document Request System, which allows residents to request and receive barangay documents online. This system greatly reduces processing time, enhances accessibility, and improves the overall efficiency of barangay operations, effectively addressing the challenges posed by traditional methods. Castagna and Bigelow (2021) define information technology as the development, processing, storage, protection, and sharing of electronic data through computers, storage, networking, and other physical devices. This move towards digitalization highlights the increasing significance of using technology to deliver timely and effective public services, in line with the broader objective of enhancing governance and community engagement at the local level.

Web-based Appointment System

According to Maribao et al. (2024), the Barangay Management System (BMS), was established to invigorate governance and provide timely services to barangays, in their view, is the most basic unit of governance in the Philippines. This system comprises various features, including those pertaining to a resident database, record-keeping, financial control, service requests, and event organization. Such features are designed to enhance workflow within administration, reduce errors, and provide efficiency in the operation of the barangay in a couple of concerns.

It explains that there are many advantages to the automated implementation of barangay processes: increased transparency and accountability, and data-backed decision-making. Thus, the system allows barangay officials to make informed decisions since data is very accurate and up-to-date and supports an evidence-based municipal governance model. There is increased interaction with the community through event management and back-and-forth engagement with citizens.

This research establishes that the integration of technology into barangay operations needs to be established not only to address inefficiencies but to further widen the scope for improving service delivery and community engagement.

Nagrama et al. (2024) stated that residents had to visit the barangay hall in person to request these documents. This process required them to fill out forms manually and wait for their requests to be processed, often leading to delays and inconvenience. However, with technological advancements and the growing demand for more efficient public services, many barangays have started to implement digital solutions to simplify administrative processes. One such innovation is the Barangay

Document Request System, which allows residents to request and receive barangay documents online. This system greatly reduces processing time, enhances accessibility, and improves the overall efficiency of barangay operations, effectively addressing the challenges posed by traditional methods.

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According to Ballaran, Donaire, Singzon, Naz, Tanael, and Centeno (2023), the Development and Evaluation of a Web-Based Barangay Profiling and Issuance System aimed to address inefficiencies in barangay governance by leveraging a centralized, web-based platform. The system aimed to make the gathering, storage, and issuance of barangay certificates and permits simpler. Regression analysis illustrates the way technology exponentiates service delivery and sharpening of decision-making processes.

Among the major cases were the lack of centralized databases in barangays, wrong data presented by the residents, and lack of proper allocation of resources. This system was developed to eradicate these problems by allowing secure data management and document processing with proper resource allocation. Evaluations apparently showed a great improvement in functional suitability, performance efficiency, and reliability of the system, which makes the platform a powerful tool in improving transparency and accountability in barangay governance. This study brings into focus the imperative need for integrating the adoption of digital solutions toward modernizing the operations of local governments and enhancing the quality of services to their constituents.

AI Chatbot

According to Pranav Nataraj Devaraj et. Al (2024), managing legal documents can be difficult due to their complexity and size. They developed an AI chatbot to make the process easier. The chatbot uses Langchain to break down large texts and find relevant information, a Flask backend to handle queries, and an Android app for user interaction. This system helps users, even those without legal expertise, access and understand legal documents more easily. The authors aim to improve it further by adding features like better design, support for more documents, and higher query limits. By responding to inquiries, providing clarifications, and supplying further resources, AI chatbots can offer prompt assistance. Additionally, chatbots can support teachers in a variety of ways by serving as virtual teaching assistants. This research aims to comprehend the whole advantages of AI chatbots in education, as well as their potential drawbacks, chances, difficulties, and future prospects (Labadze, Grigolia and Machaidze, 2024).

According to Xueliang Zhao et. Al (2019), selecting the right response in retrieval-based chatbots is challenging, especially when conversations rely on external knowledge like documents. To address this, they developed a Document-Grounded Matching Network (DGMN) that matches conversation contexts and candidate responses while grounding them in relevant documents. The model uses layers for encoding, fusing, and matching information to dynamically identify important details from both the context and the document. This approach significantly improves chatbot performance on datasets like PERSONA-CHAT and CMUDoG, outperforming previous methods and demonstrating how integrating external knowledge enhances conversational AI.

Synthesis of the Reviewed Literature and Studies

The studies highlight the growing importance of technology in local barangays. Traditional processes in barangays, such as document management and requests, are filled with inefficiencies, including delays and mismanagement. These issues hinder the accessibility of essential services that the barangay should be providing.

Digital innovations such as web-based portal document request systems can address these challenges by improving efficiency and transparency. The system uses automated processing and secured data management to reduce errors and enhance its current service.

AI-powered tools like chatbots can enhance the service by integrating the bots to the current process. This can enable efficient access to information and improve user interaction. Chatbot paired with the document management system should align with the goal of modernizing the local government to meet increasing demands for effective public service delivery.

The synthesis emphasizes that integrating technologies into local barangays not only addresses efficiency but also improves the traditional system to deliver timely, accurate and citizen-focused services.

Chapter 3

METHODOLOGY

Research Design

A thorough plan for the barangay web-based document request system was created during the system design phase. It is developed using the specifications acquired in the previous stage. This scheme contains a number of crucial features, including a program design, logical flow diagram, design, as well as the actual design and specs. According to the data flow diagram, the flow of data in the system, whereas the software design describes the general framework of the program. Furthermore, the logical design distinguishes the various parts of the system, including the database and user interface, and how they work together. Conversely, the hardware and software are defined by the physical design and specifications. prerequisites for system implementation. When combined, these elements offer a thorough plan for creating the barangay barangay web-based document request system, making sure it complies with the requirements of end users while abiding by best practices and industry norms.

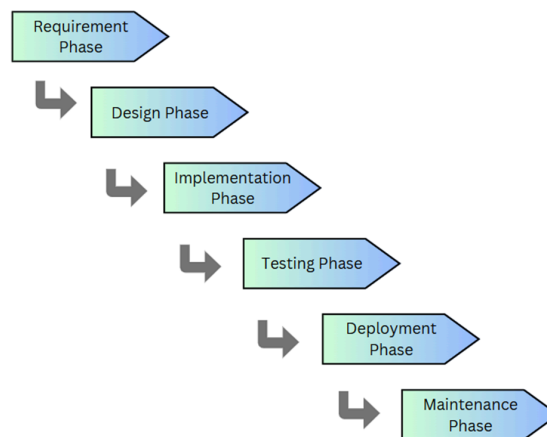
Software Development Methodology

During the implementation phase, developers will produce a system aligned with the specifications and requirements defined in the design phase. Each system component will be developed consecutively, ensuring adherence to the initial architecture and functionality. Following this, the system will move to the testing phase, where rigorous evaluations and testing will be done. These tests will ensure that the

system delivers the desired performance and service as mentioned in the requirements phase.

Once testing is complete, the system will enter the deployment phase, where it will be implemented in the barangay for end-users to make use of. Feedback will be collected to assess its initial performance.

Figure 2. Waterfall Methodology



The maintenance phase will then begin, during which developers will continue to monitor the system's performance, identify and resolve technical issues, and use the feedback gained from end-users. Improvements will be applied based on the feedback to remain an effective and user-friendly system until the system fully satisfies the needs of the users.

Sources of Data

The researchers gathered data through interviews with barangay officials from Barangay West Kamias. These interviews provided insights into the types of barangay

forms, frequently requested documents, and standard procedures for delivering services to the residents.

Additionally, the researchers distributed a survey questionnaire created through Google Forms, targeting 100 residents and 10 barangay officials of Barangay West Kamias. The survey aimed to assess the respondents' level of acceptance and evaluation of the REQWEST system.

The conduct of the survey was approved through an ethical clearance, ensuring that the data collection process adhered to ethical research standards and protected the rights and privacy of all participants.

Research Instrument

The study utilized a survey questionnaire created through Google Forms as the primary research instrument. It was designed to gather relevant data on user demographics, document request behaviors, and the level of acceptance of the REQWEST system. The questionnaire was distributed online via Facebook, specifically targeting 100 residents and 10 barangay officials of Barangay West Kamias.

It consisted of the following sections:

1. Personal Information – Collected demographic data such as gender, age, employment status, and senior/PWD classification.
2. Barangay Document Requests – Identified the most commonly requested documents from the barangay.
3. System Evaluation – Measured the level of acceptance of the REQWEST system in terms of:
 - Functional Suitability
 - Performance Efficiency

- Interaction Capability
- Reliability
- Flexibility
- Security

Responses were recorded using a Likert scale, and the data gathered was used to assess the system's usability, reliability, and effectiveness.

Data Generation Procedure

The purpose of this study is to design a simpler and more efficient barangay document request procedure in Barangay West Kamias, Quezon City. For this, the researchers designed the REQWEST system, a system to modernize and simplify this process, by making it simpler, more accurate, and easier for all. The structured approach the researchers took to collect data was used to gather the information needed to build this system.

Surveys were conducted for residents and interviews with the barangay officials in West Kamias, the researchers combined the two types of inputs to come up with insights. The surveys were strategically designed to seek to uncover the particular needs, difficulties, and patriarchy of residents in asking for documents. Interviews with barangay officials, however, the struggles faced, and what they require to make the system more effective.

Additionally, 10 barangay officials answered the same questionnaire to contribute their evaluation and feedback from an administrative standpoint. To supplement this, interviews were conducted with selected key officials, including the barangay secretary

and the barangay chairperson, to better understand current processes for document requests and service delivery.

The survey was distributed online to make sure anyone could participate. To have more meaningful conversations, the researchers spoke with barangay officials personally for the interviews. All the responses were noted carefully and analyzed and common issues and areas where improvements can be made are identified. That's the foundation for designing the REQWEST system that will help both the residents and the barangay staff make life easier.

Ethical Considerations

In this research, the data collected by face-to-face interview and online survey was conducted to the citizens of said barangay. These methods are mainly focused on how many average daily numbers of citizens are requesting documents, and a survey that shares the highest daily request in the barangay.

This research is centralized to the citizens of Barangay West Kamias District 3, Quezon City. Having nearly 4,000 residents, the sample size will be around 100 with a margin of error of 10%. Initiating this research to the barangay, researchers sought for authorization to the barangay captain. The researchers have adhered to the ethical principles of informed consent and respect for participants' privacy. A formal intent letter was made by the researchers that is passed and shown to the participants, outlining the study's purpose, methodology, and scope.

Participants were shown a formal letter of intent outlining the purpose, methodology, and scope of the research. Researchers also presented valid identification to confirm their identities and legitimize the research process. Furthermore, the study

secured ethical clearance from the Polytechnic University of the Philippines Research Ethics Committee (REC), complying with the standards set by the Philippine Health Research Ethics Board (PHREB). This clearance affirms that all documentary requirements and ethical protocols were thoroughly reviewed and approved. Participants were informed that the study aims to develop an e-service system for document requests, addressing common concerns and improving convenience for residents, particularly as the population grows and long queues become more burdensome. By upholding these ethical and professional standards, the research ensures the validity, reliability, and generalizability of its findings.

Statistical Treatment of Data

The collected raw data will undergo systematic processing, organization, and analysis to enable the formulation of inferences about the target population, the testing of hypotheses, and the identification of underlying patterns. To interpret and evaluate the data effectively, the researchers employed the following statistical methods:

1. Frequency and Percentage Distribution

To accurately assess and summarize the collected data, frequency and percentage distribution will be utilized to present the statistical proportion of each data point or category. This method involves counting the total number of observations within each data point or group and then calculating the percentage by dividing the frequency of each category by the total number of observations. This statistical treatment enables the researchers to effectively convey the relative frequency of data, particularly in analyzing the demographic profile of the respondents.

Formula:

$$\% = \left(\frac{f}{N} \right) \times 100$$

Where:

% = percentage

F = frequency

N = total number of responses

2. Mean

For symmetrical distributions, the mean serves as an effective statistical measure for determining central tendency. It is computed by summing all values within a dataset and dividing the total by the number of observations. In this study, the mean will be employed to interpret the average level of agreement on Likert scale responses, thereby assessing the perceived efficiency of the Barangay Appointment System's electronic services.

Formula:

$$\bar{x} = \frac{\sum x}{n}$$

Where:

$\bar{x}$ = mean (average)

$\sum x$ = sum of all values

N = total number of values

3. Range

The range is a statistical measure used to assess variability and describe the dispersion within a dataset. It is calculated by identifying the highest and lowest values in the dataset and subtracting the latter from the former. In this study, the range will be utilized to compare the level of agreement across Likert scale responses, providing insight into the spread of opinions between categories—specifically indicating whether responses tend to fall toward "strongly agree" or "strongly disagree."

Formula:

$$Range = x_{max} - x_{min}$$

Where:

x_{max} = Highest value in the dataset

x_{min} = Lowest value in the dataset

Likert Scale

In this capstone study, participants will be asked to assess a series of statements using a Likert scale to indicate their level of agreement or disagreement. The Likert scale will be specifically employed to evaluate resident satisfaction in West Kamias, Quezon City. The response options range from "Strongly Disagree" to "Strongly Agree," forming a 5-point scale.

Each item in the questionnaire includes predefined response choices, enabling participants to select the option that best reflects their opinion. To establish the appropriate interpretation intervals, the researchers identified the minimum and

maximum values on the scale—1 and 5, respectively. The range was calculated by subtracting the minimum value from the maximum value, yielding a difference of 4. This range was then divided by the number of points on the scale (5) to determine the interval width, resulting in an interval value of 0.80. This value was used to define the categorical boundaries for interpreting the mean scores of respondents' answers.

To effectively gather relevant information, the study will utilize six ISO quality standards—functional stability, performance efficiency, portability, reliability, usability, and security—as a framework for evaluating various aspects of the system. These standards provide a structured basis for assessing the system's overall quality and effectiveness. To ensure the validity and clarity of the questionnaire, it was reviewed and validated by Mr. Camilo P. Abogado.

Table 1
Rating and Interpretation of Questionnaires

Definition	Likert Scale	Interval
Strongly Disagree	1	1.00-1.80
Disagree	2	1.80-2.60
Neutral	3	2.60-3.40
Agree	4	3.40-4.20
Strongly Agree	5	4.20-5.00

Data Case Analysis

This study adopts a quantitative approach to guide the development of the Barangay Appointment System. Primary data were obtained through distributed forms completed by public servants, as well as questionnaires answered by officials of the barangay during the document processing phase. These primary sources provide critical information that informs both the design and implementation of the system.

In addition, the study integrates secondary data derived from both local and international research studies. These sources serve as a foundational component of the research, offering a broader and more comprehensive understanding of the subject matter. By incorporating insights from existing literature across various contexts, the secondary data enrich the interpretation of findings and contribute significantly to the overall analysis and validation of the system.

By employing a quantitative approach and integrating both primary and secondary data sources, this study aims to achieve a comprehensive and well-rounded analysis. The use of primary data, gathered directly from residents and barangay officials, provides first hand empirical evidence, while secondary data from existing local and international studies offer a broader contextual foundation. This methodological integration strengthens the reliability and validity of the research findings by ensuring that conclusions are supported by both direct observation and established scholarly knowledge.

System Architecture

Figure 3. Proposed System Architecture

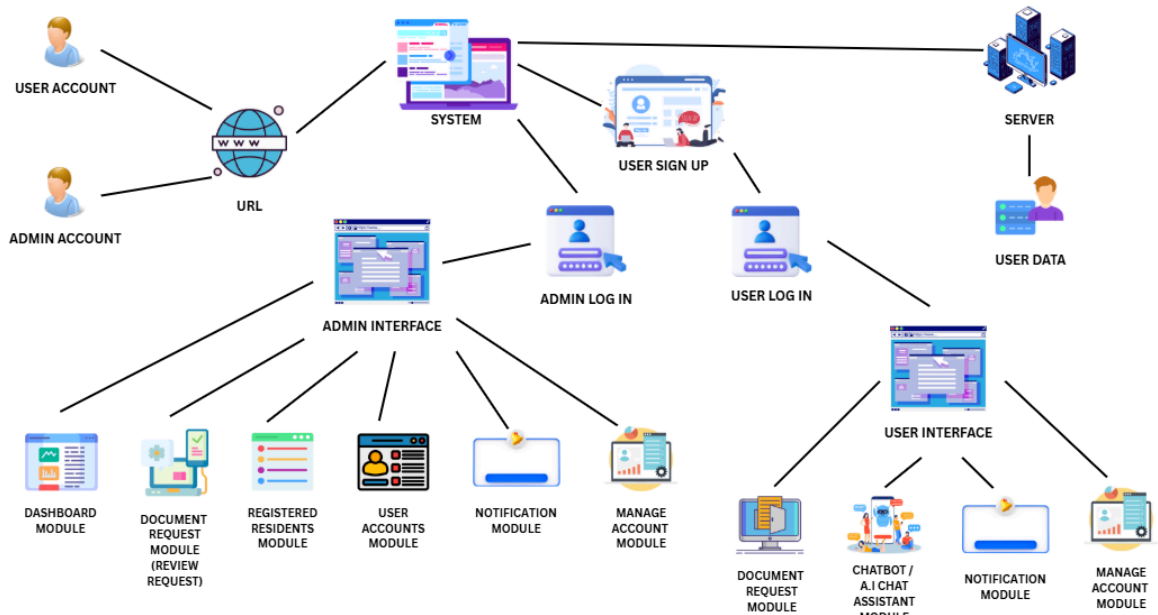


Figure 4. REQWEST User Client Flow Chart Diagram

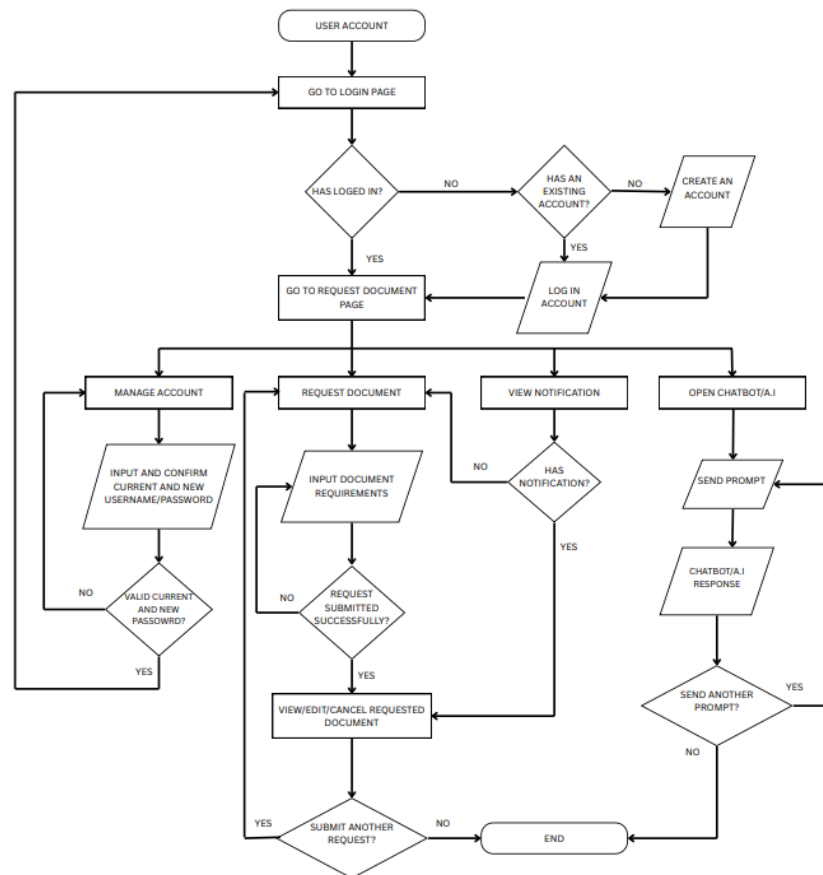


Figure 5. REQUEST Admin Client Flow Chart Diagram

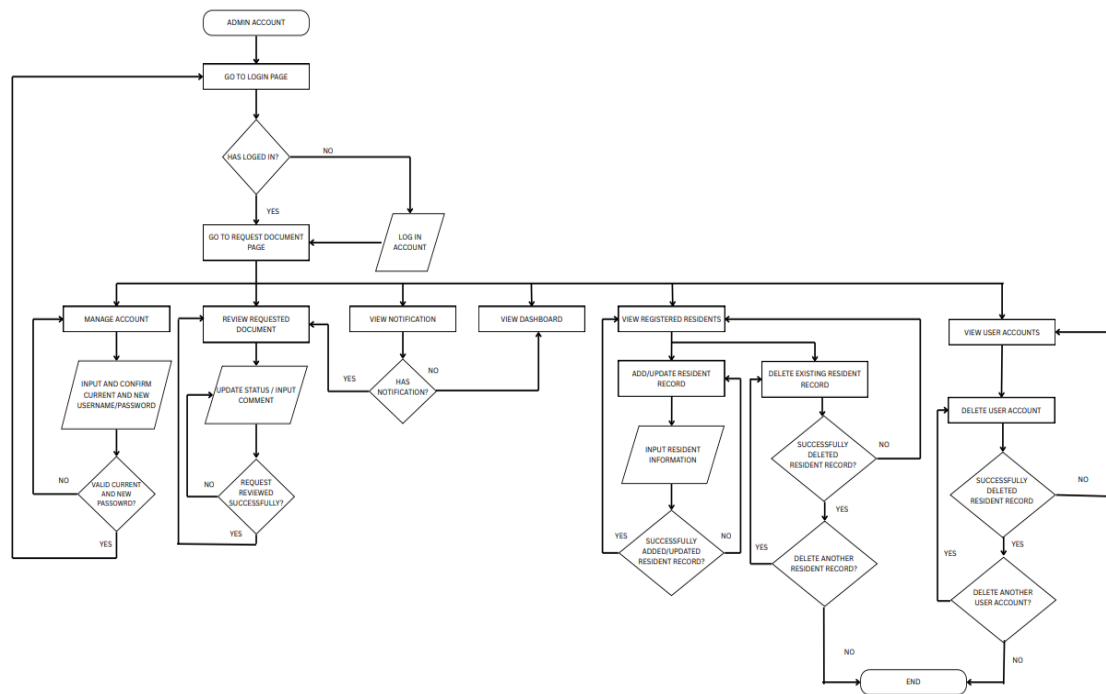
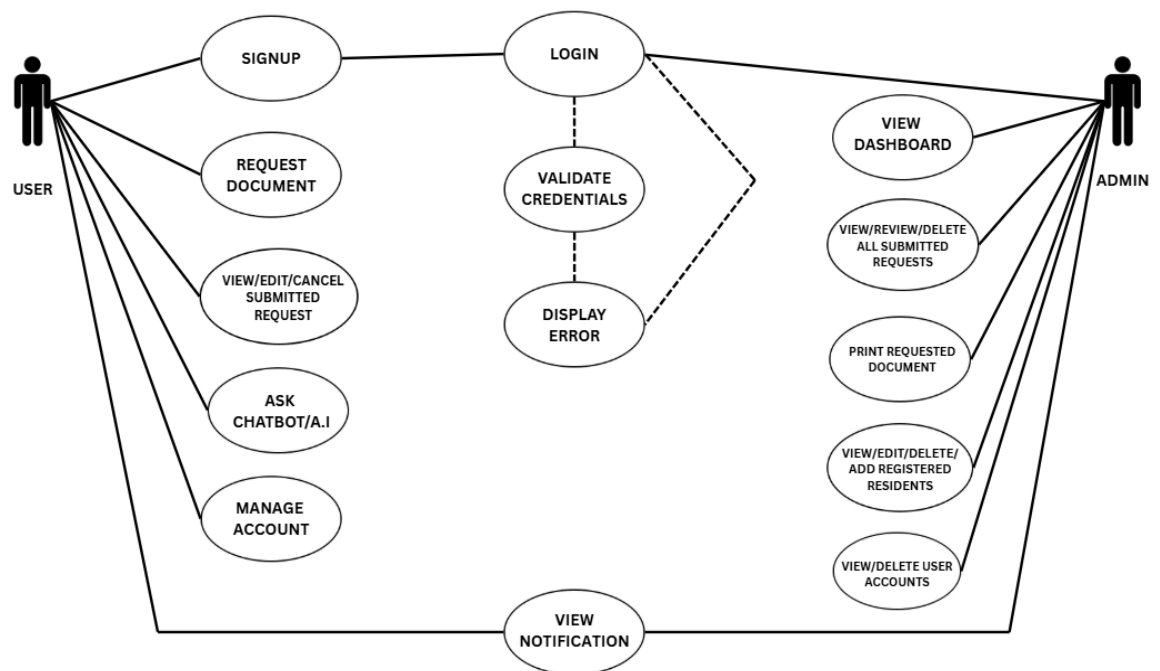


Figure 6. Use Case Diagram



Chapter 4

PRESENTATION, ANALYSIS AND RESULTS

This chapter presents a comprehensive explanation of the collected data, results of statistical analysis and the results obtained through statistical analysis, and a thoughtful interpretation of these findings based on the feedback provided by survey respondents. Tables, and descriptive narratives are used to systematically display data that directly corresponds to the research questions outlined in the problem statement, to facilitate clear understanding and comprehensive analysis.

Table 2

Hardware Components used in REQWEST

Hardware Used	Description
Development Laptop/Desktop	The REQWEST system was developed using this as the primary device. The device serves as the primary workstation for system development, testing and administrative access.
Wi-Fi Router/Modem	Provided internet connectivity for package installations, library updates, version control operations (GitHub) and remote collaboration. It provides a stable and secure internet connection which is necessary for real time system access, hosting and client server communication.
Server	This device was essential for simulating the production environment during testing and debugging. The server managed core operations such as data processing, handling form submissions, and overseeing appointment scheduling, thereby

	enabling developers to validate system performance and reliability under conditions for the system.
Storage Device	The equipment was used to strictly store files that needed a backup such as source codes, databases, as well as residents' data. The device controls efficient data management.

Table 3

Software Components used in REQUEST

Software Used	Description
Operating System (Windows 11)	The Software is useful for coding and executing the software and in the development of the REQUEST system.
Web Browser (Google Chrome)	The system was checked using a web browser for compatibility, speed and responsiveness in different web browsers to guarantee an ideal user experience for all types of devices.
Visual Studio Code (Uses HTML, CSS, JavaScript, and PHP)	REQUEST was mainly created using Visual Studio Code as the Integrated Development Environment. It provided a simple yet advanced place to write, edit and maintain code used in the system. To create the front-end, HTML was used for content and layout, CSS for style and responsiveness and JavaScript for features like form validation and interacting with users. On the server, form processing, business logic and database connections were handled using PHP. Compatibility of these languages with

	Visual Studio Code made it possible for the development team to code both the interface and the program's structure ideally.
XAMPP	The REQWEST system was built and tested using XAMPP as the host web server. It allowed the website to be run properly from the local computer. Requests from the web were managed by Apache, MySQL was used for storing the data and phpMyAdmin let users manage the tables, users and queries in the database with an interface.

Software System Analysis

The REQWEST system was implemented using a planned and structured approach. The intention of the study was to help people in barangays get documents and arrange appointments via the web. Researchers found actual difficulties such as long queues, slow processing and no clear scheduling which was acknowledged on the development.

Features like account registration, document request submission, and status tracking were created, programmed and validated for accuracy. All features improved were based on the feedback of the residents and barangay staff during the initial testing.

Detailed Development Procedures

The following general processes should be accomplished.

Research and Data Gathering

Researchers collected information initially by interviewing and surveying both the residents and officials of each barangay. The purpose is to find out the issues with manual methods for requesting documents. Most commonly mentioned issues were long lines, unexpected delays and lack of clear appointment hours. The gathered data made it clear what the system should offer and helped design solutions for users. The design of the system started with a review and summary of all the responses.

Analysis and Planning

An extensive approach to build the system is based on the identified Requirements during the gathering of the residents and Barangay officials' feedback. The aim was to design a system that would minimize the problems and delays caused by handling documents manually. From the structure of the system up to the database were planned and identified thoroughly to attain the ideal outcome of the REQWEST. System control and maintenance such as testing, debugging, and implementation were planned and maintained by the development team. Also in this phase, it was determined which tools, platforms and programming languages to use. All mentioned planning and phases are aiming to provide a dynamic and user-friendly system for the residents of Barangay West Kamias.

Programming and Development

At this stage, the researchers started to create the model using the information gathered in the previous research and planning stages. Programming tools, languages and platforms were chosen according to the findings from the analysis of the system's requirements. Data collected from the interviews with both staff and residents influenced

how the database was arranged. The developers coded the system's features and functions using best practices. Efforts to build a user-friendly interface were prioritized along with managing the website's data. The team decided on how the system would operate, how pages should be organized and what basic records are needed for both users and administrators.

Testing

The REQWEST system from its functions up to interfaces were carefully examined to ensure they worked properly and could be relied on. The developers conducted different tests to check the main features such as setting appointments, notifications, sending document requests and monitoring changes. Following the test process, the researchers ensure that the system fulfills all its objectives and carries out efficiently, aiming for a strong and dependable system for the residents of Barangay West Kamias.

Debugging

Following the testing phase, the researchers checked the code and logic of the system to deal with issues found. Debugging tools and by manually checking helped catch and correct any broken links, wrong outputs or inconsistent layouts. Methods like examining the code step by step, viewing information in the console and testing during development were helpful in finding out why there were bugs. Problems were solved by making adjustments and re-assessment to confirm the system worked correctly. Making the mentioned improvements supported the stable and strong running of the system, helping the Barangay West Kamias to provide a faster and more convenient way of digital service.

Implementation

The implementation phase focused on ensuring that the REQWEST system is ready for actual use. All significant areas of the system were finalized such as the interface, the database and the main functions. The researchers attained the system performed efficiently and able to respond accurately to user actions. Hence, the system sustains stable condition and is ready for possible use to accommodate the residents of Barangay West Kamias.

Project Investigation

This section outlines the demographic profile of the respondents who participated in the study.

Table 4
Gender of Respondents

Gender	Frequency	Percent
Female	41	41%
Male	49	49%
Prefer not to say	10	10%
Total	100	100%

Table 4 presents the gender distribution of the respondents. The majority are male, comprising 49% of the total sample, while female respondents represent the remaining 41%. The remaining 10% prefers not to say their gender. This demographic information provides insight into the gender composition of the individuals involved in the assessment of the Barangay Appointment System.

Table 5
Age of Respondents

Age	Frequency	Percent
18-24	48	48%
24-34	35	35%
35-44	7	7%
Above 45	10	10%
Total	100	100%

Based on the data presented in the table, the most represented age group among respondents who typically request documents from the barangay is 18–24 years old, comprising 48% of the total respondents. This is followed by the 25–34 age group, which accounts for 35%. In contrast, the 35–44 age group represents a smaller portion at 7%, while respondents aged 45 and above constitute only 10%. These findings suggest that younger individuals are more actively engaged in availing of barangay services, particularly in terms of document processing.

Table 6
Employment Status of the Respondents

Employment Status	Frequency	Percent
Student	41	41%
Job Seeker	8	8%
Employed	32	32%
Self-Employed	17	17%
Unemployed	2	2%
Total	100	100%

The employment status of the respondents was classified into five categories: student, job seeker, employed, self-employed, and unemployed. Out of the total sample of 100 respondents, 41% identified as students, 32% as employed, 17% as self-employed, 8% as job seekers, and only 2% as unemployed. These results indicate that the majority of the respondents are students, and unemployed comprising the smallest proportion. This distribution provides insight into the occupational background of individuals utilizing barangay services.

Table 7

Senior and PWD Respondents

Vulnerable Sectors	Frequency	Percentage
Senior	8	8%
PWD	7	7%
Non-vulnerable	85	85%
Total	100	100%

The respondent population was categorized into three groups: seniors, persons with disabilities (PWDs), and non-vulnerable individuals. Of the total 100 respondents, 8% identified as seniors, 7% as PWDs, and the remaining 85% as non-vulnerable. This distribution shows that a small portion of the population accessing barangay services belongs to vulnerable sectors, with the majority being non-vulnerable individuals.

Table 8

Most Requested Documents

Document	Frequency	Percentage
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Barangay Certificate of Indigency	29	29%
Barangay Certificate of Residency	28	28%
Barangay Permit	20	20%
Barangay Certificate of Good Moral	23	23%
Total	100	100%

This table outlines the most commonly requested documents from the barangay. The highest in demand is the Barangay Certificate of Indigency, accounting for 29% of all requests. This is followed closely by the Barangay Certificate of Residency at 28%, indicating a frequent need for residency validation in legal and government transactions. Certificates of Good Moral Character make up 23%, often required for employment or school applications, while Barangay Permits account for 20%, reflecting requests related to small business or construction permits. These results show the wide range of needs that residents approach the barangay for, with a focus on personal, academic, and business purposes.

Table 9

Respondents' Functionality Suitability Rating

Functionality	Standard Deviation	Mean	Interpretation
The system effectively processes document requests.	1.27	3.65	Agree
The system provides all necessary functions for document requests.	1.08	3.30	Neutral
The system ensures accuracy in document issuance	1.36	3.74	Agree

The system meets my expectations for processing requests.	1.22	3.68	Agree
The system reduces processing time compared to manual methods.	1.28	3.27	Neutral
General Weighted Mean:		3.53	Agree

This table displays the ratings of respondents regarding how well the system meets their functional expectations. The system's ability to accurately issue documents received a high mean score of 3.74, suggesting that users trust its precision. Similarly, the system's overall effectiveness in processing requests was rated 3.65. However, two items scored neutral: the system's capacity to provide all necessary functions (3.30) and its effectiveness in reducing processing time compared to manual methods (3.27). These suggest areas for improvement, particularly in enhancing convenience and breadth of features. The general weighted mean of 3.53 indicates moderate satisfaction with the system's functionality.

Table 10

Respondents' Performance Efficiency Rating

Efficiency	Standard Deviation	Mean	Interpretation
The system allows seamless interaction between users and the platform.	1.15	3.68	Agree
The system provides a user-friendly interface for requesting documents.	0.99	3.56	Agree
The system ensures efficient communication between residents and barangay officials.	1.12	3.43	Agree
The system minimizes errors in user inputs and system responses.	1.25	4.27	Strongly Agree
The system facilitates smooth navigation across all features.	1.08	3.79	Agree

General Weighted Mean:		3.75	Agree
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This table assesses the efficiency of the system in terms of user interaction and task execution. The highest-rated item was the system's ability to minimize errors in inputs and system responses, scoring 4.27, showing strong user confidence in system accuracy. The system's smooth navigation received a score of 3.79, while user-friendliness was also positively rated at 3.56. Lower but still positive ratings were given for efficient communication with barangay officials (3.43) and interaction flow (3.68). The general weighted mean is 3.75, indicating that the system performs well overall, though real-time communication and interaction design could still be optimized.

Table 11

Respondents' Interaction Capability Rating

Interactivity	Standard Deviation	Mean	Interpretation
The system is easy to learn, even for first-time users.	1.25	3.77	Agree
The system provides clear instructions and guidance for completing tasks.	1.06	3.79	Agree
The system allows users to efficiently navigate and operate its features.	1.13	3.56	Agree
The system minimizes errors by providing helpful prompts and feedback	1.32	4.08	Agree
The system enables users to complete tasks with minimal effort or confusion.	1.01	3.98	Agree
General Weighted Mean:		3.84	Agree

This table highlights how well users can learn and operate the system. Respondents gave high ratings to the system's ability to provide helpful prompts and

minimize errors (4.08) and to enable task completion with ease (3.98). The clarity of instructions and ease for first-time users were also rated favorably, both at 3.79 and 3.77, respectively. Navigation and operation across features scored 3.56, showing that while users find the system intuitive, further improvements in interface flow could enhance usability. The general weighted mean of 3.84 reflects a strong overall user experience.

Table 12

Respondents' Reliability Rating

Reliability	Standard Deviation	Mean	Interpretation
The system performs consistently without unexpected errors or failures.	1.27	3.65	Agree
The system reliably completes document requests without interruptions.	1.01	3.62	Agree
The system is available and accessible whenever I need to use it.	1.30	4.14	Agree
The system provides accurate and correct information in every transaction.	0.83	3.96	Agree
The system rarely experiences downtime or technical issues.	1.13	3.93	Agree
General Weighted Mean:		3.86	Agree

This table captures the system's dependability in regular operations. The system's availability and accessibility received one of the highest scores at 4.14, followed closely by its ability to deliver accurate and correct information (3.96) and consistently function without major issues (3.65). Its capacity to complete document requests without interruptions scored 3.62, while the frequency of downtime was rated

low (3.93), suggesting a generally reliable system. With a general weighted mean of 3.86, users feel confident in the system's consistency and error-free performance.

Table 13

Respondents' Flexibility Rating

Flexibility	Standard Deviation	Mean	Interpretation
The system adjusts well to different devices and screen sizes.	0.77	3.10	Neutral
The system maintains smooth performance even with high user traffic.	1.01	3.60	Agree
The system is easy to access and set up on various platforms.	0.98	3.53	Agree
The system can be adapted to future barangay needs.	1.08	3.91	Agree
The system allows integration with other digital services.	1.27	3.65	Agree
General Weighted Mean:		3.56	Agree

This table measures how adaptable the system is to different user needs and technological environments. Respondents agree that the system can be adapted to future barangay needs (3.91) and integrated with other digital services (3.65). It is also considered easy to access across various platforms (3.53) and performs well under heavy traffic (3.60). However, flexibility across different devices and screen sizes received a lower, neutral rating (3.10), pointing to a potential limitation in responsiveness. The general weighted mean of 3.56 suggests moderate system adaptability, with room for improvement in cross-device compatibility.

Table 14

Respondents' Security Rating

Security	Standard Deviation	Mean	Interpretation
The system protects my personal information from unauthorized access.	1.14	4.08	Agree
The system ensures that only authorized users can request and receive documents.	1.12	4.07	Agree
The system prevents accidental data loss or unauthorized modifications.	0.13	3.60	Agree
The system provides clear warnings or notifications before critical actions.	1.20	4.01	Agree
The system has fail-safe mechanisms to prevent data loss or unintended actions.	1.11	4.10	Agree
General Weighted Mean:		3.97	Agree

This table evaluates the security measures of the system as perceived by users. The system's ability to protect personal information and limit access to authorized users both received strong ratings of 4.08 and 4.07, respectively. Respondents also expressed confidence in the system's use of fail-safe mechanisms (4.10) and warnings before critical actions (4.01). Although slightly lower, the system's protection against accidental data loss or unauthorized changes still scored a solid 3.60. Overall, the general weighted mean of 3.97 indicates that users trust the system to safeguard their data and uphold privacy standards.

Table 15

Respondents' Summary of All Gathered Data

Functionality	Mean
The system effectively processes document requests.	3.65
The system provides all necessary functions for document requests.	3.30
The system ensures accuracy in document issuance.	3.74
The system meets my expectations for processing requests.	3.68
The system reduces processing time compared to manual methods.	3.27
Performance Efficiency	
The system allows seamless interaction between users and the platform.	3.68
The system provides a user-friendly interface for requesting documents.	3.56
The system ensures efficient communication between residents and barangay officials.	3.43
The system minimizes errors in user inputs and system responses.	4.27
The system facilitates smooth navigation across all features.	3.79
Interaction Capability	
The system is easy to learn, even for first-time users.	3.77
The system provides clear instructions and guidance for completing tasks.	3.79
The system allows users to efficiently navigate and operate its features.	3.56
The system minimizes errors by providing helpful prompts and feedback.	4.08
The system enables users to complete tasks with minimal effort or confusion.	3.98
Reliability	
The system performs consistently without unexpected errors or failures.	3.65
The system reliably completes document requests without interruptions.	3.62

The system is available and accessible whenever I need to use it.	4.14
The system provides accurate and correct information in every transaction.	3.96
The system rarely experiences downtime or technical issues.	3.93
Flexibility	
The system adjusts well to different devices and screen sizes.	3.10
The system maintains smooth performance even with high user traffic.	3.60
The system is easy to access and set up on various platforms.	3.53
The system can be adapted to future barangay needs.	3.91
The system allows integration with other digital services.	3.65
Security	
The system protects my personal information from unauthorized access.	4.08
The system ensures that only authorized users can request and receive documents.	4.07
The system prevents accidental data loss or unauthorized modifications.	3.60
The system provides clear warnings or notifications before critical actions.	4.01
The system has fail-safe mechanisms to prevent data loss or unintended actions.	4.10

The survey indicates an overall average mean score of 3.75, which corresponds to an "agree" rating in terms of verbal interpretation. This suggests that respondents generally have a positive perception of the system's overall performance. They found it functional, efficient, user-friendly, reliable, flexible, and secure. While most aspects were rated favorably, the results also highlight certain areas such as functionality breadth and device adaptability where further improvements could enhance user satisfaction and system effectiveness.

Table 16
Gender of Admins

Gender	Frequency	Percent
Female	3	30%
Male	7	70%
Prefer not to say	0	0%
Total	10	100%

This table presents the gender distribution among the barangay administrators who participated in the study. The majority are male (70%), while females represent 30%. No respondents chose to withhold their gender. This data provides a basic demographic profile of the admin respondents, indicating a male-dominated group involved in barangay operations.

Table 17
Barangay Position of Admins

Position	Frequency	Percent
Barangay Chairman	1	10%
Secretary	1	10%
SK Chairman	1	10%
SK Secretary	1	10%
SK Kagawad	6	60%
Total	10	100%

This table lists the positions held by the admin respondents within their respective barangays. The most common position is SK Kagawad, representing 60% of

the group. Other positions include Barangay Chairman, Secretary, SK Chairman, and SK Secretary, each at 10%. This distribution reflects a diverse set of roles, with a strong representation from the Sangguniang Kabataan sector.

Table 18

Most Requested Documents as per Admins

Document	Frequency	Percentage
Barangay Certificate of Indigency	5	50%
Barangay Certificate of Residency	2	20%
Barangay Permit	2	20%
Barangay Certificate of Good Moral	1	10%
Total	10	100%

This table summarizes the types of documents most frequently requested by residents, according to the admin respondents. The Barangay Certificate of Indigency is reported as the most requested document (50%), followed by both the Certificate of Residency and Barangay Permit (20% each). The Certificate of Good Moral is the least cited, at 10%. These insights align with data from resident respondents and highlight the system's relevance to frequently needed documents.

Table 19

Admins' Functionality Suitability Rating

Functionality	Standard Deviation	Mean	Interpretation
The system effectively processes document requests.	0.52	4.60	Strongly Agree

The system provides all necessary functions for document requests.	0.52	4.40	Strongly Agree
The system ensures accuracy in document issuance	0.70	4.60	Strongly Agree
The system meets my expectations for processing requests.	0.68	4.30	Strongly Agree
The system reduces processing time compared to manual methods.	0.53	4.50	Strongly Agree
General Weighted Mean:		4.48	Strongly Agree

This table presents the administrators' evaluation of the system's functional suitability. High mean scores were recorded across all items, with the system's effectiveness in processing document requests and accuracy in document issuance both receiving 4.60. The lowest but still strong rating of 4.30 was for meeting expectations. The general weighted mean of 4.48 suggests that admins strongly agree on the system's functional reliability and usefulness.

Table 20

Admins' Performance Efficiency Rating

Efficiency	Standard Deviation	Mean	Interpretation
The system allows seamless interaction between users and the platform.	0.52	4.60	Strongly Agree
The system provides a user-friendly interface for requesting documents.	0.53	4.50	Strongly Agree
The system ensures efficient communication between residents and barangay officials.	0.48	4.70	Strongly Agree
The system minimizes errors in user inputs and system responses.	0.52	4.60	Strongly Agree

The system facilitates smooth navigation across all features.	0.68	4.30	Strongly Agree
General Weighted Mean:		4.54	Strongly Agree

This table assesses the efficiency of the system based on admin responses. The highest mean of 4.70 was given to the system's ability to ensure communication between residents and officials, indicating strong support for its communication features. All other items scored between 4.30 and 4.60, contributing to a general weighted mean of 4.54, which indicates that administrators perceive the system as highly efficient and responsive to user actions.

Table 21

Admins' Interaction Capability Rating

Interactivity	Standard Deviation	Mean	Interpretation
The system is easy to learn, even for first-time users.	0.71	4.50	Strongly Agree
The system provides clear instructions and guidance for completing tasks.	0.82	4.00	Agree
The system allows users to efficiently navigate and operate its features.	0.70	4.40	Strongly Agree
The system minimizes errors by providing helpful prompts and feedback	0.52	4.40	Strongly Agree
The system enables users to complete tasks with minimal effort or confusion.	0.52	4.60	Strongly Agree
General Weighted Mean:		4.38	Strongly Agree

This table evaluates how well the system supports user interaction, based on admin feedback. The highest rating (4.60) was given to ease of task completion, while

the lowest score (4.00), though still positive, was for the clarity of instructions. The rest of the items fall between 4.40 and 4.50, resulting in a general weighted mean of 4.38. This indicates strong agreement that the system is user-friendly and intuitive even for first-time users.

Table 22

Admins' Reliability Rating

Reliability	Standard Deviation	Mean	Interpretation
The system performs consistently without unexpected errors or failures.	0.52	4.40	Strongly Agree
The system reliably completes document requests without interruptions.	0.84	4.40	Strongly Agree
The system is available and accessible whenever I need to use it.	0.53	4.50	Strongly Agree
The system provides accurate and correct information in every transaction.	0.53	4.50	Strongly Agree
The system rarely experiences downtime or technical issues.	0.68	4.30	Strongly Agree
General Weighted Mean:		4.42	Strongly Agree

This table shows how the system performs in terms of reliability according to admins. All items received high agreement scores, with system availability and accuracy of information both rated at 4.50. Even the lowest item, downtime experience, received a favorable 4.30. The general weighted mean of 4.42 reflects consistent reliability and uninterrupted system performance.

Table 23

Admins' Flexibility Rating

Flexibility	Standard Deviation	Mean	Interpretation
The system adjusts well to different devices and screen sizes.	1.16	3.30	Neutral
The system maintains smooth performance even with high user traffic.	0.82	4.00	Agree
The system is easy to access and set up on various platforms.	0.53	4.50	Strongly Agree
The system can be adapted to future barangay needs.	0.88	4.10	Agree
The system allows integration with other digital services.	0.53	4.50	Strongly Agree
General Weighted Mean:		4.08	Agree

This table reports on how adaptable the system is to different platforms and future use. The highest-rated items, ease of access on various platforms and integration with other services, each received a strong 4.50. The lowest score of 3.30 was given to the system's adjustment across different screen sizes, suggesting room for improvement in interface responsiveness. The general weighted mean of 4.08 reflects overall agreement that the system is flexible and forward-compatible.

Table 24

Admins' Security Rating

Security	Standard Deviation	Mean	Interpretation
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The system protects my personal information from unauthorized access.	0.52	4.40	Strongly Agree
The system ensures that only authorized users can request and receive documents.	0.53	4.50	Strongly Agree
The system prevents accidental data loss or unauthorized modifications.	0.82	4.30	Strongly Agree
The system provides clear warnings or notifications before critical actions.	0.48	4.30	Strongly Agree
The system has fail-safe mechanisms to prevent data loss or unintended actions.	0.53	4.50	Strongly Agree
General Weighted Mean:		4.40	Strongly Agree

This table reflects the admins' perception of the system's data protection features. High scores were given across all items, with fail-safe mechanisms and access control both rated at 4.50. The lowest rating of 4.30 was for data loss prevention and critical warnings still within the "strongly agree" range. The general weighted mean of 4.40 indicates high trust in the system's ability to maintain data privacy and safeguard user information.

Table 25

Admins' Summary of All Gathered Data

Functionality	Mean
The system effectively processes document requests.	4.60
The system provides all necessary functions for document requests.	4.40
The system ensures accuracy in document issuance.	4.60
The system meets my expectations for processing requests.	4.30
The system reduces processing time compared to manual methods.	4.50
Performance Efficiency	

The system allows seamless interaction between users and the platform.	4.60
The system provides a user-friendly interface for requesting documents.	4.50
The system ensures efficient communication between residents and barangay officials.	4.70
The system minimizes errors in user inputs and system responses.	4.60
The system facilitates smooth navigation across all features.	4.30
Interaction Capability	
The system is easy to learn, even for first-time users.	4.50
The system provides clear instructions and guidance for completing tasks.	4.00
The system allows users to efficiently navigate and operate its features.	4.40
The system minimizes errors by providing helpful prompts and feedback.	4.40
The system enables users to complete tasks with minimal effort or confusion.	4.60
Reliability	
The system performs consistently without unexpected errors or failures.	4.40
The system reliably completes document requests without interruptions.	4.40
The system is available and accessible whenever I need to use it.	4.50
The system provides accurate and correct information in every transaction.	4.50
The system rarely experiences downtime or technical issues.	4.30
Flexibility	
The system adjusts well to different devices and screen sizes.	3.30
The system maintains smooth performance even with high user traffic.	4.00

The system is easy to access and set up on various platforms.	4.50
The system can be adapted to future barangay needs.	4.10
The system allows integration with other digital services.	4.50
Security	
The system protects my personal information from unauthorized access.	4.40
The system ensures that only authorized users can request and receive documents.	4.50
The system prevents accidental data loss or unauthorized modifications.	4.30
The system provides clear warnings or notifications before critical actions.	4.30
The system has fail-safe mechanisms to prevent data loss or unintended actions.	4.50

This table provides a comprehensive summary of the average ratings given by admin respondents across all evaluation categories: functionality, performance efficiency, interaction capability, reliability, flexibility, and security. The highest scores were observed in Performance Efficiency and Security, where most items received mean ratings ranging from 4.30 to 4.70, indicating strong agreement that the system operates smoothly, securely, and effectively. In contrast, the lowest-rated item was the system's adaptability to different devices and screen sizes, with a mean of 3.30, suggesting a need for improvements in mobile or responsive design. Overall, the average scores reflect that the barangay administrators strongly agree with the system's effectiveness, user-friendliness, and dependability, while also recognizing a few areas that could benefit from future development.

Chapter 5

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

Summary

Barangay West Kamias currently relies on a fully manual process for scheduling appointments and issuing official documents such as barangay clearances. This practice is slow, labor-intensive, and prone to errors, leading to long queues, limited transparency, and restricted access for residents. To address these inefficiencies, the study proposes the development of a digital Barangay Document Request System aimed at modernizing and streamlining these essential services.

The objectives of this proposed system include optimizing efficiency by reducing processing times for appointments and document releases, modernizing operations through the transition from paper-based workflows to a digital platform, and improving transparency and accessibility by enabling residents to monitor the status of their requests in real time. Furthermore, the system is designed to enhance governance by equipping barangay officials with accurate, centralized data that supports responsive decision-making.

To evaluate the current system and gather relevant data, the researchers conducted a survey among registered voters in Barangay West Kamias. Descriptive statistics were employed to interpret the survey results. Frequency and percentage distribution were used to determine how often specific opinions were expressed, while the mean provided a measure of overall agreement on statements related to system efficiency using a Likert scale. Additionally, the range was calculated to assess response variability, highlighting areas of strong consensus or divergence.

The findings revealed strong resident support for a digital system, with many respondents anticipating faster transactions and reduced congestion at the barangay hall. The survey's mean scores indicated high agreement (exceeding 4.0 on a 5-point scale) regarding the benefits of an online platform in improving convenience and transparency. Moreover, low range values within the “strongly agree” responses reflected a high level of consistency among participants.

The Barangay Document Request System is envisioned as a centralized web application where residents can book appointment slots online, submit and track requests for barangay clearances, IDs, and related documents, and receive notifications about the status and release schedules of their requests. For barangay staff, the system will offer an integrated backend dashboard that automates scheduling, consolidates resident records, and provides real-time analytics.

The anticipated benefits of the system include improved efficiency through reduced wait times and faster processing, data-driven governance enabled by immediate insights into service volumes and community needs, enhanced transparency and accountability through clear audit trails and status updates, and citizen-centric service delivery by offering anytime, anywhere access to barangay services.

In conclusion, implementing the Barangay Document Request System will transform Barangay West Kamias into a model of responsive, technology-enabled local governance. By adding manual procedures with a streamlined digital workflow, the barangay stands to significantly improve service quality, strengthen public trust, and set a benchmark for innovation in neighboring communities.

Findings

A survey conducted among 100 participants revealed notable demographic and usage patterns within the sample population. The gender distribution indicated a disparity, with males comprising 49% of respondents and females accounting for 41%. The majority of document requesters were within the 18–24 age group, representing 48% of the sample. In terms of employment status, students constituted the largest group at 41%, followed by employees at 32%, which included 5% self-employed, and job seekers at 8%. Regarding the types of documents requested, Barangay Indigency was the most commonly sought, with 29% of respondents requesting it. This was followed by Barangay Residency at 28%, Barangay Good Moral at 23%, and Barangay Permit at 20%.

The findings indicate that the majority of barangay document requesters are young adults aged 18 to 24. Students represent the largest group among the respondents, reflecting the high demand for barangay services within this demographic. Barangay Indigency and Residency documents were the most commonly requested, suggesting that social and residential verification are primary reasons for document requests. Furthermore, the notable number of requests for Barangay Good Moral and Permit documents points to active engagement in academic and community activities. These results imply that the barangay primarily serves young residents who require official documents for educational and community purposes.

The survey findings reveal a moderately high level of satisfaction among respondents concerning the system's functionality, efficiency, usability, portability, reliability, and security. The system received positive ratings across all evaluated aspects, with an overall mean score of 3.75, corresponding to an interpretation of

"Agree," which indicates a moderately high level of user satisfaction. However, flexibility received the lowest average score among the criteria. This lower rating may be attributed to the current absence of a mobile-friendly interface, limiting users' ability to access the system conveniently across different devices. Despite the generally positive feedback, there remains a continuous effort to enhance the system to better address evolving user needs and expectations. Identifying areas for improvement, particularly in flexibility, will contribute to the sustained advancement of the system's performance and user experience.

Conclusions

1. The researchers uncovered several key findings among the participants. In terms of gender distribution, the majority of respondents are male, comprising 49% of the total. Female participants account for 41%, while 10% chose not to disclose their gender. This reflects a noticeable gender imbalance among the population. As for age demographics, individuals aged 18 to 24 represent the largest group requesting documents from the barangay, making up 48% of the respondents. Looking into the employment background of the participants, the data shows that students form the largest portion, comprising 41% of the surveyed group. This highlights how barangay west kamias is utilized by the younger, school-attending population. When looking at the vulnerable sectors, the data implies that access to these services is more common among non-vulnerable individuals, who make up 85% of the population, compared to seniors and those with disabilities. In addition, the Barangay Certificate of Indigency stands out as the most commonly requested document, with 29% of participants stating they required it.

2. The researchers conclude that in terms of functionality, efficiency, interactivity, reliability, and flexibility, the system received favorable ratings, especially in minimizing errors and being consistently available for use. The functionality section for the respondents received an average rating of 3.52, while the admin received an average rating of 4.48 indicating that the system performed in line with user expectations. Likewise, the efficiency category also scored 3.75 for the respondents and 4.54 for the admin, reflecting the system's smooth and effective operation. In terms of the Interactivity the data from the respondents earned a general weighted mean of 3.84, while the data from the admins has 4.38, showing users' confidence in the system's interactivity measures. Reliability received an average score of 3.86 from the respondents and 4.42 from the admins, demonstrating users' trust in the system's ability to securely store data. For flexibility, the respondents gave the system an average of 3.56, while the admins gave 4.08, suggesting that users encountered minimal limitations when adapting the system to different needs or environments. The security component received a weighted mean of 3.97 from the respondents, while 4.40 from the admins, suggesting that users felt confident in the system's protective capabilities. Among all the aspects evaluated, flexibility received the lowest average score, highlighting the need to enhance the system's adaptability to various user needs and environments. Despite generally positive feedback and strong agreement on its effectiveness, it remains important to continually strive for improvements in this area.
3. The researchers conclude that the implementation of the REQUEST system significantly improved services in Barangay West Kamias for both administrators and citizens. The system streamlined document processing and reduced manual

workload for the barangay officials, while citizens benefited from the convenience of online requests, avoiding the hassle of visiting the barangay in person. It addressed common issues such as poor communication, delays, misunderstandings about requirements, and inefficient data handling. The system's two-way notification feature kept both users and staff updated, speeding up transactions. The integration of an AI-powered chatbot added innovation by assisting with basic inquiries. System security was also prioritized; only registered users and authorized barangay staff can access specific features, ensuring that personal information and sensitive records remain protected. These improvements demonstrate the system's effectiveness in increasing efficiency and public satisfaction.

4. The researchers conclude that while the system was designed with user accessibility and convenience in mind, it currently lacks a dedicated mobile interface. This absence of mobile responsiveness limits the usability of the system by residents who rely on smartphones. Despite the lack of responsiveness, the system provides improved accessibility and convenience when accessed through desktop or laptop computers. Its clear layout and streamlined document request process contribute to a more efficient user experience compared to traditional methods. However, the absence of mobile compatibility remains a limitation. To extend the system's usability and reach, especially for residents who rely on smartphones, future enhancements should focus on developing a mobile-friendly interface.

Recommendations

1. Implementation of multi-language support: Adding a language selection feature to REQWEST will allow users to navigate the system in their preferred language, making it more inclusive and user-friendly for individuals in multicultural barangay communities.
2. Responsive design for various devices: The system should be optimized for mobile phones, tablets, and other smart devices to ensure accessibility and usability across different screen sizes, improving user experience on the go.
3. Development of a mobile application: Creating a dedicated mobile application for REQWEST will offer easier access and real-time notification support, which will greatly benefit users needing to request and track documents remotely.
4. Expansion to support additional document types: While REQWEST currently handles requests for Certificate of Indigency, Certificate of Residency, Barangay Permit, and Barangay Good Moral Certificate, it is recommended to include support for other possible barangay-related documents in the future, such as Barangay Business Clearance, Endorsement Letters, or other special certifications as needed by residents.
5. Enhancement of security features: Strengthening system security such as encryption, two-factor authentication, and secure user data storage should be a top priority to protect against unauthorized access, data breaches, and fraud.
6. Improvement of user interface design: Enhancing the overall design, layout, and visual appeal of the system will create a more pleasant experience for users and encourage continuous use of REQWEST.

By following these recommendations, REQWEST will be able to improve its accessibility, functionality, security, and user satisfaction. Ultimately providing a more efficient and inclusive document request platform for barangay-level services.

APPENDICES

Appendix 1: Survey Questionnaire

Polytechnic University of the Philippines Institute of Technology

Dear Respondent,

We are third-year Information Technology students from the Polytechnic University of the Philippines – Institute of Technology. As part of our research, we are developing REQWEST, a web-based application designed to enhance the monitoring and information system for requesting documents at West Kamias, Quezon City.

This survey aims to gather insights on:

- Challenges associated with the current system for managing and requesting barangay documents.
- Potential solutions to improve the efficiency of the document requesting process.
- User experience in navigating the REQWEST System, evaluated based on ISO 25010 software quality standards.

Your participation is entirely voluntary, and all responses will be kept confidential and used solely for research purposes. Your feedback is invaluable in refining the system to ensure efficiency, usability, and reliability for students, faculty, and partner Barangay West Kamias, Quezon City

We sincerely appreciate your time and valuable input.

Best regards,

Alonsagay, Mark Kieth M.

Baugbog, Warren A.

Caponga, CeeJhay E.

Ganotice, Alen Jerru C.
Mapa, Lester James C.
Querubin, Ronald Bong V.
Tan, Germain A.

Information Technology 3-2
Polytechnic University of the Philippines – Institute of Technology

CONSENT OF AGREEMENT

In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), we value and respect your privacy. By participating in this survey, you agree to the following terms:

1. The information you provide will be used solely for research purposes related to the development and evaluation of the REQWEST system for West Kamias.
2. Your responses will remain confidential and anonymous. No personal identifiers will be collected or shared without your explicit consent.
3. The collected data will be securely stored and accessed only by the research team for analysis.
4. Participation in this survey is voluntary, and you may choose to withdraw at any time without any consequences.
5. By proceeding with this survey, you acknowledge that you have read and understood this consent agreement and voluntarily agree to participate.

Part I. PERSONAL INFORMATION

This section collects basic personal information to help categorize responses and ensure the relevance of the data collected. Your information will remain confidential and will only be used for research purposes related to the development of the document request system.

1. Name (Optional)
2. Gender:
☐ Male
☐ Female
☐ Rather not say
3. Age:
☐ 18-24
☐ 25-34
☐ 35-44
☐ 45 and Above
4. Senior/PWD Status:
☐ Senior Citizen
☐ PWD
5. What is your current employment status?
☐ Student
☐ Job Seeker
☐ Employed
☐ Self-Employed
☐ Unemployed
6. Frequently Requested Barangay Documents

- _____ Barangay Certificate of Indigency
- _____ Barangay Certificate of Residency
- _____ Barangay Permit
- _____ Barangay Certificate of Good Moral

Part II. SYSTEM REVIEW

In this section, you will evaluate the REQUEST prototype based on selected **ISO 25010** software quality characteristics. This evaluation will focus on the following six characteristics: **functional suitability, performance efficiency, interaction capability, reliability, flexibility, and safety.**

Using the scale given below, select the number that describes the quality of the system.

5 - Strongly Agree 4 - Agree 3 - Slightly Agree/Disagree 2 - Disagree 1 - Strongly Disagree

Functional Suitability	5	4	3	2	1
The system effectively processes document requests.					
The system provides all necessary functions for document requests.					
The system ensures accuracy in document issuance.					
The system meets my expectations for processing requests.					
The system reduces processing time compared to manual methods.					

Performance Efficiency	5	4	3	2	1
The system allows seamless interaction between users and the platform.					
The system provides a user-friendly interface for requesting documents.					
The system ensures efficient communication between residents and barangay officials.					
The system minimizes errors in user inputs and system responses.					

The system facilitates smooth navigation across all features.					
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Interaction Capability	5	4	3	2	1
The system is easy to learn, even for first-time users.					
The system provides clear instructions and guidance for completing tasks.					
The system allows users to efficiently navigate and operate its features.					
The system minimizes errors by providing helpful prompts and feedback.					
The system enables users to complete tasks with minimal effort or confusion.					

Reliability	5	4	3	2	1
The system performs consistently without unexpected errors or failures.					
The system reliably completes document requests without interruptions.					
The system is available and accessible whenever I need to use it.					
The system provides accurate and correct information in every transaction.					
The system rarely experiences downtime or technical issues.					

Flexibility	5	4	3	2	1
The system adjusts well to different devices and screen sizes.					
The system maintains smooth performance even with high user traffic.					
The system is easy to access and set up on various platforms.					
The system can be adapted to future barangay needs.					
The system allows integration with other digital services.					

Safety	5	4	3	2	1
The system protects my personal information from unauthorized access.					
The system ensures that only authorized users can request and receive documents.					
The system prevents accidental data loss or unauthorized modifications.					
The system provides clear warnings or notifications before critical actions.					
The system has fail-safe mechanisms to prevent data loss or unintended actions.					

Admin Survey

Polytechnic University of the Philippines Institute of Technology

Dear Respondent,

We are third-year Information Technology students from the Polytechnic University of the Philippines – Institute of Technology. As part of our research, we are developing REQWEST, a web-based application designed to enhance the monitoring and information system for requesting documents at West Kamias, Quezon City.

This survey aims to gather insights on:

- Challenges associated with the current system for managing and requesting barangay documents.
- Potential solutions to improve the efficiency of the document requesting process.
- User experience in navigating the REQWEST System, evaluated based on ISO 25010 software quality standards.

Your participation is entirely voluntary, and all responses will be kept confidential and used solely for research purposes. Your feedback is invaluable in refining the system to ensure efficiency, usability, and reliability for students, faculty, and partner Barangay West Kamias, Quezon City

We sincerely appreciate your time and valuable input.

Best regards,

Alonsagay, Mark Kieth M.
Baugbog, Warren A.
Caponga, CeeJhay E.
Ganotice, Alen Jerru C.
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Querubin, Ronald Bong V.

Tan, Germain A.

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Polytechnic University of the Philippines – Institute of Technology

CONSENT OF AGREEMENT

In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), we value and respect your privacy. By participating in this survey, you agree to the following terms:

1. The information you provide will be used solely for research purposes related to the development and evaluation of the REQWEST system for West Kamias.
2. Your responses will remain confidential and anonymous. No personal identifiers will be collected or shared without your explicit consent.
3. The collected data will be securely stored and accessed only by the research team for analysis.
4. Participation in this survey is voluntary, and you may choose to withdraw at any time without any consequences.
5. By proceeding with this survey, you acknowledge that you have read and understood this consent agreement and voluntarily agree to participate.

Part I. PERSONAL INFORMATION

This section collects basic personal information to help categorize responses and ensure the relevance of the data collected. Your information will remain confidential and will only be used for research purposes related to the development of the document request system.

1. Name (Optional)
2. Barangay Position:
3. Gender:
☐ Male
☐ Female
☐ Rather not say
4. Frequently Requested Barangay Documents (Base on your experience, which of the following document (currently available in REQWEST) is the most frequently requested by residents?)
☐ Barangay Certificate of Indigency
☐ Barangay Certificate of Residency
☐ Barangay Permit
☐ Barangay Certificate of Good Moral

Functional Suitability	5	4	3	2	1
The system effectively allows us to process document requests submitted by residents.					
The system provides all the necessary administrative functions for managing document requests (e.g., verification, approval, issuance).					
The system ensures accuracy in generating and issuing documents to residents.					
The system meets our expectations for managing and fulfilling document requests.					
The system reduces the administrative burden of processing document requests compared to manual methods.					

Performance Efficiency	5	4	3	2	1
The system allows for efficient management of resident requests.					
The system provides a clear and efficient interface for barangay officials to manage requests.					
The system ensures timely processing and completion of document requests.					
The system minimizes errors in administrative actions and system outputs.					
The system allows for quick access to relevant information for processing requests.					

Interaction Capability	5	4	3	2	1
The system is easy for barangay officials to learn and use for managing requests.					
The system provides clear instructions and guidance for administrative tasks.					
The system allows barangay officials to efficiently navigate and operate its administrative features.					

The system minimizes administrative errors through helpful prompts and feedback.					
The system enables barangay officials to manage requests with minimal effort or confusion.					

Reliability	5	4	3	2	1
The system performs consistently and allows me to manage document requests without errors.					
The system reliably processes and records resident requests without data loss or failure.					
The system is available and responsive whenever I need to access the admin panel..					
The system provides accurate data and updates in real-time.					
The system rarely requires troubleshooting or technical support intervention.					

Flexibility	5	4	3	2	1
The system adapts well to different administrative workflows or procedures.					
The system maintains stable performance even with a high volume of administrative tasks or requests.					
The system is easy for administrators to access and use on different devices (if applicable).					
The system can be adapted to future administrative needs of the barangay.					
The system allows for potential integration with other barangay management tools.					

Security	5	4	3	2	1
The system effectively protects sensitive resident data from unauthorized administrative access.					

The system ensures that only authorized barangay officials can manage and access request information.					
The system prevents accidental data loss or unauthorized administrative modifications.					
The system provides clear warnings or notifications before critical administrative actions.					
The system has fail-safe mechanisms to prevent data loss or unintended administrative actions.					